

FIZZA RUBAB

✉ rubabfiz@msu.edu

🔗 [fizza-rubab.github.io](https://github.com/fizza-rubab)

🌐 [fizzarubab](https://fizzarubab.com)

🎧 [FizzaRubab](https://www.instagram.com/FizzaRubab)

EDUCATION

Michigan State University

Aug 2024 - Present

PhD Computer Science and Engineering

East Lansing, US

- CGPA: 4.00/4.00
- Award: University Distinguished Fellowship
- Research Area: Computer Graphics/Computer Vision/3D Deep Learning

Habib University

Aug 2019 - May 2023

BS Computer Science - Double Minor in Electrical Engineering and Mathematics

Karachi, PK

- CGPA: 3.99/4.00, Rank: 1/114

EXPERIENCE

Mach9

May 2026 - Present

Perception Intern

San Francisco, USA

- Researching neural architectures and representations for **vectorized 3D HD map construction** from LiDAR point clouds and urban imagery.
- Implemented **cyclic permutation-invariant matching** in a transformer decoder, enabling accurate closed-loop polyline prediction for traffic islands and curb rings.

Michigan State University

Aug 2024 - May 2026

Research Assistant

East Lansing, USA

- Conducting research on **neural network-based geometry processing and modelling** under [Prof. Yiyong Tong](#)
- Designed and implemented an interactive neural shape editing framework for neural signed distance functions. Introduced tubular sampling and a stroke-based formulation, achieving versatile edits and up to a **15×** speedup over prior methods. Published in **High Performance Graphics 2025**
- Currently exploring gradient-domain processing techniques for alternate shape representations such as Geometry Images

Max Planck Institute for Informatics

Jan 2024 - July 2024

Visiting Research Intern

Saarbrücken, DE

- Conducted research on **neural integration** under the supervision of [Dr. Thomas Leimkühler](#)
- Researched neural field-based approaches for learning repeated antiderivatives of multidimensional signals. Designed a **taxonomy of methods** for learning integrals supported by thorough experimental evaluation and analysis. Published and presented at **Vision Modelling and Visualization 2025**
- Applied learned integrals of motion signals, images, and 3D SDFs for signal filtering and image-based lighting

Wolfram Research

Nov 2023 - May 2024

Machine Learning Intern

Remote

- **Audited Wolfram Language's machine learning functionality** against algorithms and concepts from standard ML textbooks and libraries and identified missing capabilities
- Implemented missing classification and regression metrics (Huber loss, Jaccard score, KL divergence) and evaluation visualizations (lift curves, calibration plots)
- Developed a **parameterized random data generator** for systematic testing of classification and clustering algorithms

Farnevo

Aug 2023 - Dec 2023

Machine Learning Engineer

Karachi, PK

- Developed object detection based pipelines using YoloV8, DeTR, MaskRCNN models to improve crop detection on drone images of the field
- Implemented a **semi-supervised auto-labeling pipeline** for plant detection which improved accuracy by **+15%**
- Developed gap filling prescription algorithm which identified gaps in row crops up to **85%** accurately in dense fields

Wolfram Summer School

Jun 2023 - Jul 2023

Research Participant

Waltham, USA

- Developed a **monocular 3D human pose estimation pipeline** by combining CenterNet-based 2D keypoint detection with MiDaS monocular depth estimation
- Reconstructed and interactively visualized 3D human skeletons from RGB images in the Wolfram Language
- Published the results in the [Wolfram Summer School 2023 Proceedings](#)

National Resource for Network Biology, Google Summer of Code

Jun 2022 - May 2023

Google Summer of Code Contributor

Remote

- Worked with [Prof. Pedro Mendes](#) and [Dr. Hasan Baig](#) on **Cloud-COPASI**, a biological analysis platform
- Redesigned Cloud-COPASI to **streamline HPC job submissions** without requiring complex SSH configuration
- Implemented **COPASI compute pools on AWS Cloud**, enabling non-HPC affiliated users to run large simulations

PUBLICATIONS

1. **F. Rubab**, Y. Tong and A. Ross, "Are Face Embeddings Compatible Across Deep Neural Network Models?" in arXiv, 2026
2. **F. Rubab** and Y. Tong, "Interactive Stroke-based Neural SDF Sculpting," in High Performance Graphics (HPG), 2025
3. **F. Rubab**, N. Nsambi, M. Balint, F. Mujkanovic, H. Seidel, T. Ritschel and T. Leimkühler, "Learning Neural Antiderivatives," in Vision Modeling and Visualization (VMV), 2025
4. I. Siddiqui, **F. Rubab**, H. Siddiqui and A. Samad, "Poet Attribution of Urdu Ghazals using Deep Learning," in 3rd International Conference on Artificial Intelligence (ICAI), 2023
5. **F. Rubab**, S. Hafeez, M. H. Qazi, M. H. Syed, B. I. Azeemi and A. J. Kayani, "A Comparative Survey of Solutions to Russel's Paradox," in 4th International Conference on Computing, Mathematics and Engineering Technologies (ICOMET), 2023
6. S. Hafeez, S. Pervez, R. Naeem, **F. Rubab** and S. Raza, "Prompt Emergency Response: PLSP Mobile Application for Effective Cardiac Arrest and Trauma Management in Pakistan" in 13th IEEE Global Humanitarian Technology Conference (GHTC), 2023

AWARDS AND HONORS

Phi Kappa Phi Membership Nation's oldest and largest all-discipline honors society membership based on excellence	Michigan State University, 2026
Tau Beta Pi Membership National engineering honor society membership for academic and professional excellence	Michigan State University, 2025
University Distinguished Fellowship Prestigious five-year fellowship awarded to 40 exceptional incoming graduate students	Michigan State University, 2024
Chancellor's Yohsin Medal For graduating with highest honor of excellent academics and metacurriculars	Habib University, 2023
Dean's Medal For graduating with the highest CGPA in Computer Science Program	Habib University, 2023
Best Research Award For producing the best research publication in Computer Science	Habib University, 2023
Dean's Honor List For being one of the top 10% students in the program each semester	Habib University, 2020-2022
President's Honor List For maintaining position on Dean's Honor List in consecutive semesters	Habib University, 2021-2022
High Academic Achievement Scholarship Additional 10% Scholarship for the Top 3 students in the school each semester	Habib University, 2020-2023
2nd Position in Speed Programming, CyberSecurity Hackathon Awarded by president of Pakistan for securing first runner up position in speed programming	Ignite, 2021
Bronze Medalist in ICPC Regionals For excellent performance in International Collegiate Programming Contest, Topi Regionals	ICPC, 2022
HU-TOPS Merit Scholarship 100% scholarship for four years awarded to top 50 students from all over Pakistan	Habib University, 2019-2023
First Position in National Examinations Securing highest grade nationwide with distinctions in mathematics and chemistry	AKUEB, 2019

ACADEMIC SERVICES

- **Organizer/Demonstrator** for Girls Math and Science Day, MSU, 2026
- **Teaching Assistant at Michigan State University** for CSE335 Object-oriented Software Development, 2025-2026
- **Section Instructor, Stanford University** for CS106A Code in Place course in 2023 and 2024 (remote)
- **Teaching Assistant at Habib University for 6 courses:** Natural Language Processing, Linear Algebra, Nature of Computation, Discrete Mathematics, Object Oriented Programming and Computer Science Freshman Seminar
- **Programming Fundamentals Workshop Facilitator** for incoming computer science class of 2025, Habib University
- **Programming Bootcamp Instructor** for middle school students at Aga Khan School

SKILLS

Programming: Python, C/C++, OpenGL, Java, MATLAB, C#, Mathematica
ML Tools: Numpy, Pytorch, Jax, Pandas, Scikit-Learn, Tensorflow, OpenCV